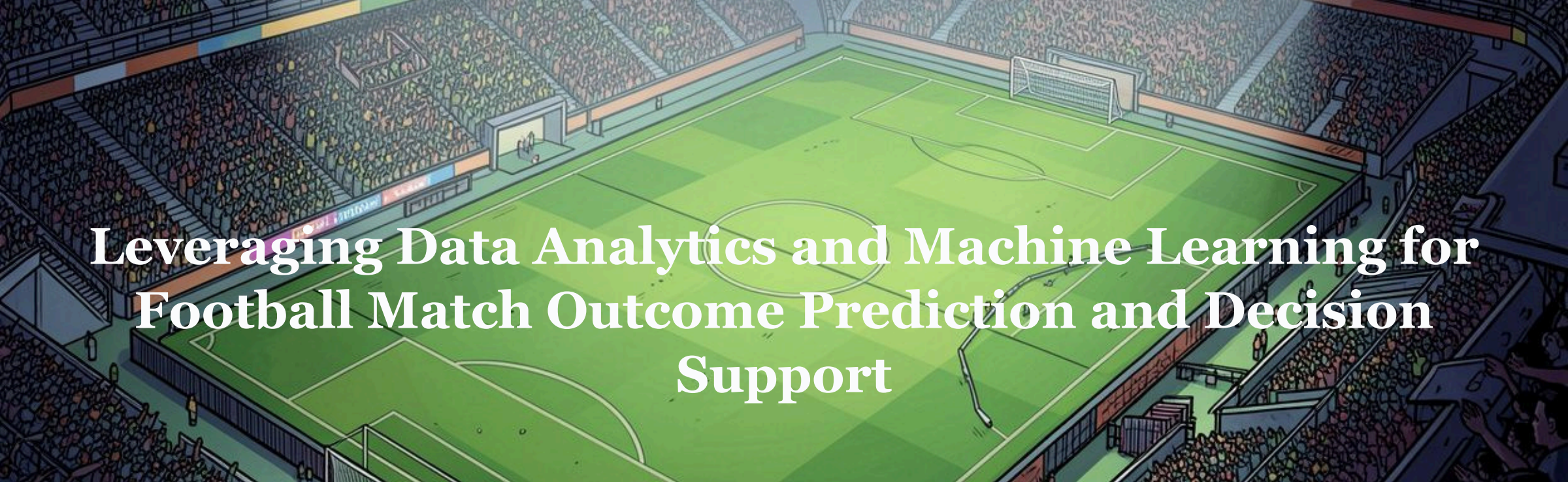




Leveraging Data Analytics and Machine Learning for Football Match Outcome Prediction and Decision Support

Field of study: Advanced Analytics - Big Data

Vishnu Gopal - 140100

Supervision of Dr. Zbigniew Gontar

Research Problem

Football match outcomes are inherently uncertain and influenced by multiple interacting factors. Historical football data is increasingly available, but is often fragmented and not directly transformed into actionable decision-support insights. Many advanced approaches depend on detailed event or tracking data that can be proprietary, expensive, and computationally demanding.

The Core Challenge

Accessible historical match-level data exists but is rarely transformed into meaningful predictive features and interpretable machine-learning predictions for practical decision support.

Research Focus

The thesis investigates whether accessible historical match-level data can be transformed into meaningful predictive features and interpretable ML predictions — seeking a balance between **predictive performance, interpretability, and practical usability**.



Aim, Objectives & Research Questions

Aim & Objective

Design and implement a data analytics and machine-learning system that predicts football match outcomes and provides analytical insights for decision support.

- Design an end-to-end analytical pipeline
- Transform historical records into ML-ready indicators
 - Compare three supervised-learning approaches
- Evaluate predictive performance using multiple metrics
- Provide interpretable outputs for football decision-making
 - Keep the framework modular, reproducible, and extensible

Research Questions

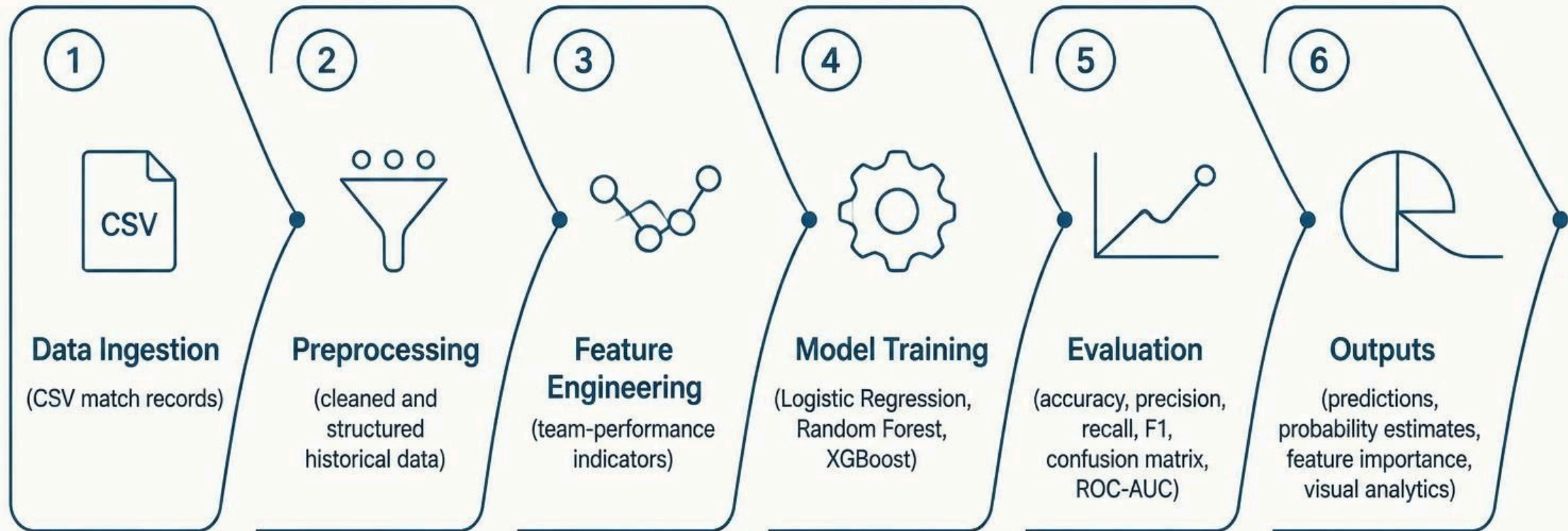
- 1 How can a data analytics and ML-based system be designed and implemented to predict football match outcomes and support decision-making?
- 2 What features derived from historical match data most influence match outcome predictions?
- 3 How do Logistic Regression, Random Forest, and XGBoost compare in predictive performance?
- 4 How can predictive insights be used to support decision-making in sports management?

Data Foundation

Accessible, structured match-level data obtained from Kaggle forms the backbone of this study. The dataset uses aggregated match-level records rather than detailed event or tracking data, ensuring public accessibility and reproducibility.

Attribute	Description
Source	Publicly available football data obtained from Kaggle
Attribute	Structured historical match-level records
Boundary	Home team, away team, goals scored, and match outcomes
Granularity	Aggregated match-level data rather than detailed event/tracking data
Processing	Local Python environment using standard analytics libraries
Scalabilty	Architecture designed for future larger datasets and distributed processing

Analytical Framework & Architecture



Data Processing & Preprocessing

Reparing historical records for predictive modelling requires a disciplined, chronologically aware pipeline. Each step ensures data quality, temporal integrity, and readiness for machine learning.

Attribute	Description
. Data ingestion	Load structured CSV match data.
Cleaning	Ensure consistency and usable variables.
Chronological processing	Preserve historical ordering.
Feature generation	Build team-level performance indicators
ML dataset	Construct model-ready observations.
Train/test split	20% hold-out testing strategy

Feature Engineering

Historical match records are transformed into predictive indicators that capture attacking effectiveness, defensive performance, competitive consistency, and short- term momentum. Chronological design ensures only information available before the predicted match is used, preventing data leakage.

Historical Measure	Predictive Indicator	What It Captures
Average goals scored	Attacking effectiveness	Offensive output per match
Average goals conceded	Defensive performance	Defensive vulnerability
Home / away win rate	Competitive consistency	Winning frequency by venue
Goal difference strength	Relative attacking strength	Comparative goal differential
Win-rate difference	Relative competitive strength	Comparative win rates
Defensive difference	Relative defensive capability	Comparative defensive record
Recent form	Short-term performance / momentum	Rolling 5-match performance
Form difference	Relative recent performance	Comparative momentum

❏ **Recent form** is calculated as rolling performance over the five most recent historical matches, capturing short-term momentum. **Comparative features** directly represent relative team strength between home and away sides.

Feature Engineering — Core Equations

Comparative indicators represent relative team strength by computing the difference between home and away team metrics. Chronological feature generation prevents data leakage and simulates realistic pre-match information.

Goal Difference Strength

$$home_avg_goals - away_avg_goals$$

Win-Rate Difference

$$home_win_rate - away_win_rate$$

Defensive Difference

$$home_avg_conceded - away_avg_conceded$$

Form Difference

$$home_recent_form - away_recent_form$$

Machine-Learning Models

Three complementary supervised-learning approaches are compared under identical feature sets and evaluation conditions. Logistic Regression serves as the interpretable baseline; Random Forest adds non-linear ensemble learning with feature importance; XGBoost provides a boosting-based comparison with higher model complexity.

Logistic Regression

Interpretable baseline — Probability-based, simple, and transparent. Provides a clear probabilistic output and serves as the reference model for interpretability.

Random Forest

Ensemble comparison — Captures non-linear relationships and produces feature importance scores, offering insight into which variables drive predictions.

XGBoost

Boosting comparison — Sequential gradient boosting with higher model complexity, typically delivering strong predictive performance on structured tabular data.

Evaluation Methodology

All three models are evaluated under the same feature set and testing conditions. A 20% hold-out testing subset with fixed random state ensures reproducibility. **Target:** 1 = Home Win; 0 = Draw or Away Win.

Metric	Definition
Accuracy	Overall proportion correctly classified
Precision	Reliability of positive (home-win) predictions
Recall	Ability to identify actual home wins
F1-Score	Balance between precision and recall
Confusion Matrix	Distribution of correct and incorrect classifications
ROC-AUC	Discrimination across classification thresholds

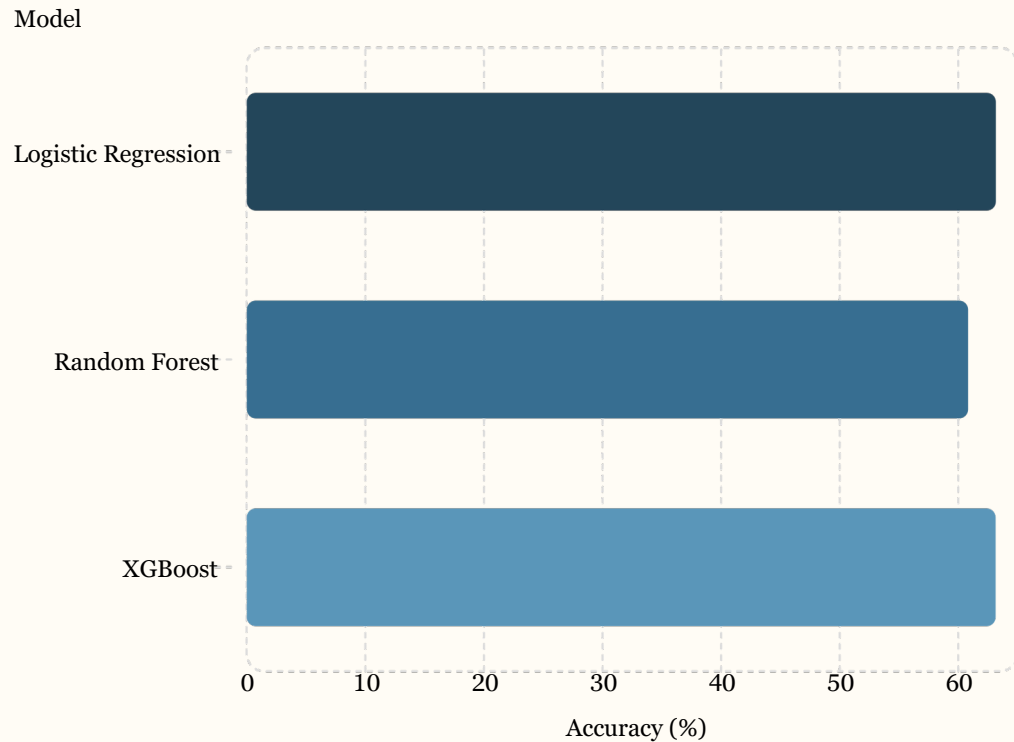
Fair Comparison	Reproducibility	Decision Support
Same engineered features and identical evaluation conditions across all three models.	Fixed random state and 20% hold-out test split ensure consistent, repeatable results.	Outputs include predictions, probability estimates, feature importance, and visual analytics.

Evaluation Metrics — Mathematical View

Accuracy alone is not sufficient. A complete evaluation requires multiple complementary metrics.

Metric	Formula
Accuracy	$(T P + T N) / (T P + T N + F P + F N)$
Precision	$T P / (T P + F P)$
Recall	$T P / (T P + F N)$
F1-score	$2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$

Model Performance — Testing Accuracy



Key Findings

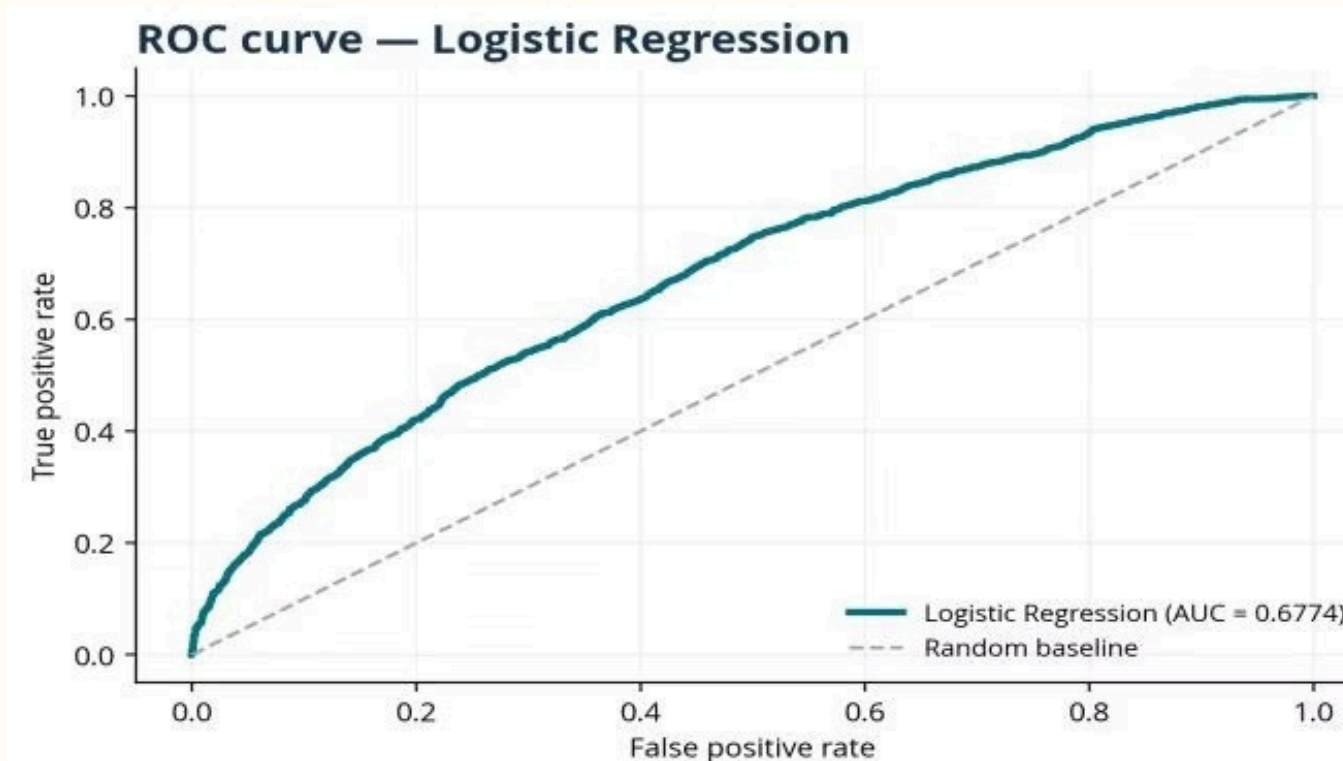
- **Logistic Regression** led at 63.17% — the highest
- accuracy **XGBoost** was comparable at 63.15%
- **Random Forest** was competitive at 60.83%
- Greater model complexity did *not* automatically improve accuracy

Classification Behavior — Confusion Matrices

Results on the 20% hold-out test set reveal each model's discrimination patterns.

Model	Actual 0 → Pred 0	Actual 0 → Pred1	Actual 1 → Pred 1
Logistic Regression	2,007	703	1147
Random Forest	1892	818	1145
XGBoost	2063	647	1090

ROC Curve — Logistic Regression



$AUC = 0.6774$

The teal curve stays consistently above the random baseline, confirming the model performs better than chance.

False Positive Rate

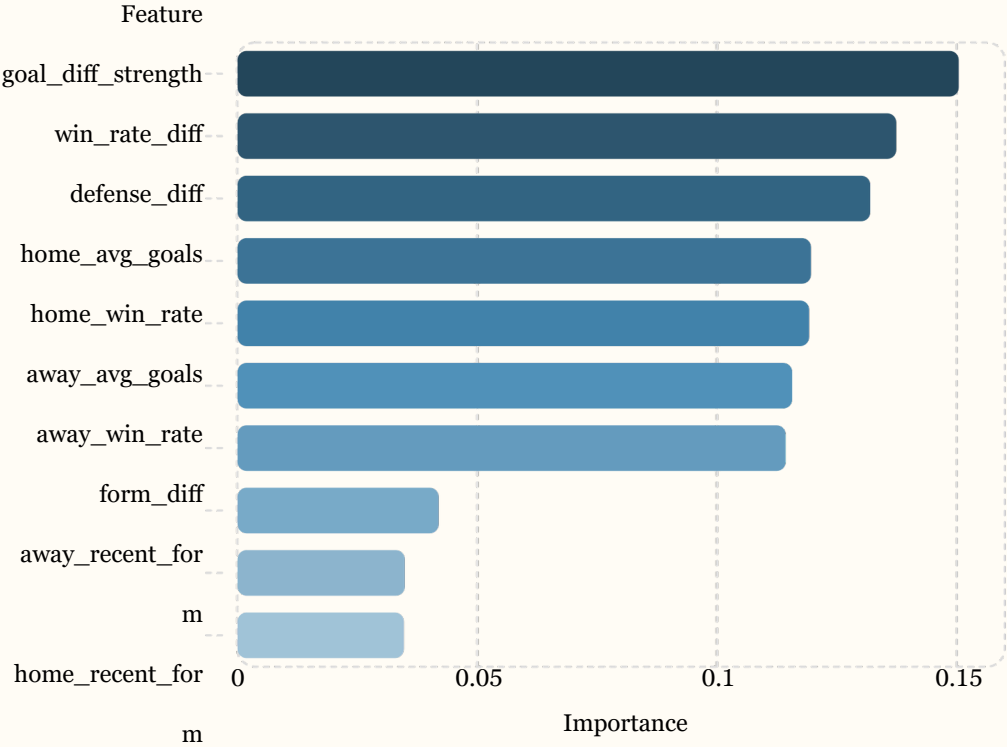
X-axis, 0.0 to 1.0

True Positive Rate

Y-axis, 0.0 to 1.0

Feature Importance — Random Forest

Which engineered features matter most for predicting match outcomes?



Top Drivers

- **goal_diff_strength** (0.150) — strongest
- predictor **win_rate_diff** (0.137) —
- comparative form **defense_diff** (0.132) —

defensive gap
Recent form features ranked lowest, suggesting longer-term indicators carry more signal.



Interpretation & Decision Support

Logistic Regression

Highest accuracy (63.17%); probability outputs add nuance beyond binary labels.

Random Forest

Competitive accuracy with interpretable feature importance for opponent analysis.

XGBoost

Comparable performance (63.15%) but no accuracy gain over simpler models.

- ❑ **Predictive results complement expert judgement — they do not replace it.**

Practical Implications

Direct relevance to sports analytics and management.



Match Preparation

Historical strength, recent form, and probability estimates complement pre-match analysis.



Opponent Analysis

Comparative features highlight attacking, defensive, and competitive differences.



Performance Evaluation

Feature importance identifies variables most associated with predictions.



Strategic Planning

Predictive insights support coaching and management discussions with evidence.



Accessible Analytics

Public data and standard Python tools reduce dependence on proprietary infrastructure.



Scalability

Modular design can accommodate richer datasets, dashboards, and distributed processing.

Limitations & Future Development

Current Limitations

- Aggregated match-level data only — no tactical or spatial detail Binary target groups draws and away wins, reducing granularity Excludes injuries, weather, squad rotation, and managerial
- changes
- Hold-out validation only; k-fold cross-validation not
- implemented

Future Direction

- Integrate event-level and tracking data
- Add contextual variables: injuries, weather, tactics
- Evaluate deep learning and transformer-based sequence models Explore real-time deployment and Apache Spark scalability Develop interactive dashboards for broader decision support

Conclusions

Reproducible Pipeline

Historical football data transformed into a clean, ML-ready dataset with chronological feature engineering that prevents data leakage.

Logistic Regression Wins

Highest accuracy (63.17%); Random Forest competitive with interpretability; XGBoost comparable but no improvement.

Interpretability Matters

Structured data + engineered features + interpretable ML = useful predictive and decision-support insights.

- ❏ **Key Takeaway:** A reproducible, interpretable analytics pipeline can transform accessible historical football data into predictive insights that support evidence-based decision-making.

